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Costly punishment sustains indirect reciprocity under low defection detectability

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ABSTRACT

Cooperation is fundamental to human societies, and indirect reciprocity, where individuals cooperate to build a positive reputation for future benefits, plays a key role in promoting it. Previous theoretical and experimental studies have explored both the effectiveness and limitations of costly punishment in sustaining cooperation. While empirical observations show that costly punishment by third parties is common, some theoretical models suggest it may not be effective in the context of indirect reciprocity, raising doubts about its potential to enhance cooperation. In this study, we theoretically investigate the conditions under which costly punishment is effective. Building on a previous model, we introduce a new type of error in perceiving actions, where defection may be mistakenly perceived as cooperation. This extension models a realistic scenario where defectors have a strong incentive to disguise their defection as cooperation. Our analysis reveals that when defection is difficult to detect, norms involving costly punishment can emerge as the most efficient evolutionarily stable strategies. These findings demonstrate that costly punishment can play a crucial role in promoting cooperation within indirect reciprocity.

1. Introduction

Cooperation is a fundamental aspect of human societies, enabling individuals to work together and achieve outcomes that would be difficult or impossible to accomplish alone (Rand and Nowak, 2012; Melis and Semmann, 2010). One of the key mechanisms that sustains cooperation is indirect reciprocity, where individuals gain a reputation based on their behavior, influencing their future interactions with others (Nowak and Sigmund, 2005; Wedekind and Braithwaite, 2002; Sigmund, 2012; Santos et al., 2021; Okada, 2020). In this framework, those who cooperate tend to receive a good reputation, which encourages others to cooperate with them in subsequent interactions. Conversely, individuals who defect acquire a bad reputation, which can lead to a long-term loss in future interactions.

People often use punishment as a strategy to deter defection and promote cooperation. By imposing penalties on defectors, groups can maintain or even enhance cooperation. Numerous behavioral experiments and real-world observations show that costly punishment — where individuals incur personal costs to penalize defectors — is a common phenomenon in human societies (Fehr and Gächter, 2002; De Quervain et al., 2004; Rockenbach and Milinski, 2006; Mathew and Boyd, 2011; Wu et al., 2022; Balafoutas et al., 2014; Ule et al., 2009; Kanakogi et al., 2022). People often punish perceived wrongdoers even at a cost to themselves, reflecting a deep-rooted willingness to uphold social norms.

Although costly punishment is often regarded as a mechanism to promote cooperation in various contexts (Boyd and Richerson, 1992; Henrich and Boyd, 2001; Boyd et al., 2003; Brandt et al., 2006; Liu et al., 2018, 2019), some theoretical studies question the effectiveness of punishment in promoting cooperation (Guala, 2012; Wu et al., 2022; Raihani and Bshary, 2015, 2019; Li et al., 2018; Hauser et al., 2014; Rand and Nowak, 2011). Ohtsuki et al. (2009) have suggested that the effectiveness of costly punishment may be limited in the context of indirect reciprocity. Their study indicates that the introduction of costly punishment often results in lower overall payoffs compared to scenarios where punishment is not used. Moreover, in environments that are too noisy or where the cost of punishment is excessively high, costly punishment can become less efficient than strategies like always defecting (ALLD). As a result, the conditions under which costly punishment is effective appear to be quite narrow. This suggests that simple defection (withholding cooperation from freeriding) is sufficient to deter freeriding, making costly punishment unnecessary for sustaining cooperation. However, their model assumes that players' actions are always perfectly observed, which is often unrealistic in real-world scenarios. This raises the critical question of whether the detectability of defection itself may influence the effectiveness of costly punishment in indirect reciprocity.

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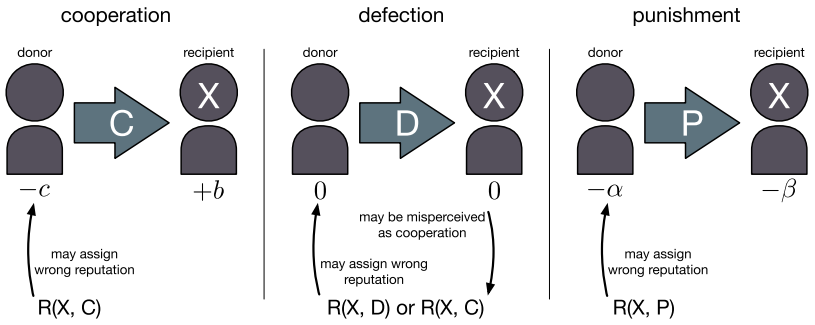


Fig. 1. A schematic diagram of the model. At each step, a donor and a recipient are randomly selected. The donor then chooses an action from three options: cooperation (C), defection (D), or punishment (P). If the donor chooses to cooperate, the recipient receives a benefit b, while the donor incurs a cost c. In the case of defection, both players receive a payoff of zero. If the donor opts for punishment, the recipient incurs a loss β, and the donor pays a punishment cost α. The donor's action influences their reputation as perceived by the recipient. The new reputation depends on the recipient's current reputation X and the donor's action. If the donor defects, there is a probability ϵ_{DC} that the action will be misperceived as cooperation. In all other cases, the action is correctly identified. Additionally, the assessment process is prone to errors. With a probability μ, the opposite reputation is assigned to the donor.

Table 1
Well-known examples of social norms without costly punishment. The action rule prescribes which action to take when the recipient is assessed as good (G) or bad (B). The assessment rule $R(X, A)$ prescribes the probability of assigning a good reputation to the donor, depending on the recipient's reputation X and the donor's action A. An asterisk ($*$) indicates that the value can be arbitrary. Simple Standing and Stern Judging norms are the only ESS norms that achieve full cooperation in the low-error limit $\mu \to 1$. These are two of the "leading eight" norms.

Name	Action rule		Assessment rule			
	$S(G)$	$S(B)$	$R(G,C)$	$R(G,D)$	$R(B,C)$	$R(B,D)$
Image Scoring	C	D	1	0	1	0
Simple Standing	C	D	1	0	1	1
Stern Juddging	C	D	1	0	0	1
ALLC	C	C	$*$	$*$	$*$	$*$
ALLD	D	D	$*$	$*$	$*$	$*$

Our study addresses this question by exploring how costly punishment fosters cooperation when defection is difficult to detect. Defectors have a natural incentive to disguise their actions as cooperative, making it difficult to differentiate between genuine cooperation and defection (Kiyonari, 2010). Conversely, cooperators strive to ensure their behavior is accurately recognized to avoid unfair punishment. This creates an asymmetry: while cooperation is often observable, defection may go unnoticed or be misinterpreted.

To incorporate this asymmetry in imperfect monitoring, we extend the standard model of Ohtsuki et al. (2009) by introducing the concept of the "detectability of defection", a parameter that quantifies the ease of identifying defection accurately. Our theoretical analysis reveals that social norms incorporating costly punishment emerge as the most efficient evolutionarily stable strategies under a broad range of conditions, particularly when defection is hard to detect. When defection detectability is low, maintaining cooperation becomes more challenging, but costly punishment helps sustain cooperation by countering undetected defection.

These findings contrast with previous studies that focused on errors in distinguishing good and bad reputations (Ohtsuki et al., 2009; Nakamura and Masuda, 2011; Ohtsuki and Iwasa, 2004, 2006; Murase and Hilbe, 2023). In our model, the challenge lies in distinguishing defection from cooperation, leading to a starkly different outcome. This highlights the importance of considering the nature of errors when evaluating the role of costly punishment in indirect reciprocity.

2. Model

In this study, we follow the basic framework established by Ohtsuki et al. (2009). We consider an infinitely large population of players, where individuals repeatedly interact in pairwise donation games. In each round, two players are randomly chosen, one as the donor and the other as the recipient. The donor chooses an action from cooperation (C), defection (D), or punishment (P). Cooperation incurs a cost $c > 0$ to the donor and provides a benefit $b > c$ to the recipient. Defection results in a payoff of zero for both players. Punishment incurs a cost $\alpha > 0$ to the donor and imposes a loss of $\beta > 0$ on the recipient. If the donation game is played only once, the donor benefits more by defecting, creating a social dilemma. However, we consider the scenario where members of the population repeatedly engage in many donation games with changing donor-recipient pairs. During these interactions, players build reputations based on their past actions, which may influence future cooperation. Following the standard convention in public assessment models (Ohtsuki and Iwasa, 2004, 2006; Murase and Hilbe, 2023), we assume that reputations are binary, with players being either good (G) or bad (B), and that each player's reputation is publicly shared without disagreement.

How players' reputations are formed and how they act based on these reputations depend on their social norms. In our study, a social norm consists of an action rule S and an assessment rule R, as illustrated in Fig. 1. The action rule $S(X)$ determines the action—either C, D, or P—when acting as a donor. This choice may depend on the recipient's reputation $X \in \{G, B\}$. The action rule is defined as a map $S : X \to A$, where $X \in \{G, B\}$ represents the recipient's reputation and $A \in \{C, D, P\}$ represents the donor's action. Additionally, we consider an assessment rule $R(X, A)$, which determines the probability of assigning a good reputation to the donor based on the donor's action and the recipient's reputation. In this study, we consider only deterministic norms, so R is constrained to be either 0 or 1; that is, $R : X \times A \to \{0, 1\}$. We consider the so-called "second-order" social norms, where the assessment rules depend on both the recipient's reputation and the donor's action. As a reference, we list several well-known norms without costly punishment in Table 1. Hereafter, we denote a norm with $S(G) = C$ and $S(B) = D$ as a CD norm, a norm with $S(G) = C$ and $S(B) = P$ as a CP norm, and so on.

Reputation dynamics are subject to errors. In our model, we consider two types of errors: assessment errors and detection errors. An assessment error, also conventionally introduced in previous studies, may occur when a new reputation is assigned to the donor. With probability μ ($\leq 1/2$), the opposite reputation is assigned to the donor. This can equivalently be interpreted as the population's ability to distinguish between good and bad reputations with probability $q = 1 - 2\mu$ (Ohtsuki et al., 2009). The parameter q quantifies the ability to distinguish between good and bad reputations and is called the 'social resolution'.

The second type of error is detection error, which is newly introduced in this study. Actions are not always correctly observed and may be misperceived. Here, we consider the case where defection is perceived as cooperation, due to the natural incentive for defectors

to disguise their defection as cooperation. When a donor defects, the action is perceived as cooperation with probability ϵ_{DC} and correctly identified as defection with probability $1-\epsilon_{DC}$. We define the detectability of defection as $\xi = 1-\epsilon_{DC}$. When $\epsilon_{DC} = 0$ ($\xi = 1$), defections are perfectly detected, corresponding to the baseline model. When $\epsilon_{DC} = 1$ ($\xi = 0$), defections are indistinguishable from cooperation and are always misperceived as cooperation.

While it is straightforward to consider other types of errors, such as execution errors where the donor's intended action is not carried out correctly, we focus on these two types to keep the model as simple as possible. These errors underscore the role of the costly punishment in sustaining cooperation through indirect reciprocity, as shown in the following analysis. We will discuss the types of errors under which costly punishment can be effective in Section 4.

Given these two types of errors, the effective assessment rule $\tilde{R}(X,A)$ (Murase and Hilbe, 2023) is expressed as follows:

$$\begin{cases} \tilde{R}(X,C) &= (1-2\mu)R(X,C)+\mu \\ \tilde{R}(X,D) &= (1-2\mu)[\epsilon_{DC}R(X,C)+(1-\epsilon_{DC})R(X,D)]+\mu \\ \tilde{R}(X,P) &= (1-2\mu)R(X,P)+\mu \end{cases} \tag{1}$$

For any X and A, $\mu \le \tilde{R}(X,A) \le 1-\mu$, ensuring that the average reputation converges to a unique equilibrium value regardless of the initial state when $\mu > 0$ (Ohtsuki and Iwasa, 2004, 2006; Murase and Hilbe, 2023).

All subsequent analysis reproduces the findings of the baseline model when $\xi = 1$.

3. Analysis

In the following, we derive the norms that are evolutionarily stable strategies (ESS) within the model described above. Among these ESS norms, we identify the most efficient ones — those leading to the highest average payoff at equilibrium — for the given environmental parameters $\{b, c, \alpha, \beta, q, \xi\}$.

ESS norms are defined as those that satisfy the following condition: A single mutant player (one with a different action rule S') has a strictly lower payoff than the players following the resident action rule S for all $S' \neq S$. Obviously, any DD norms (ALLD), are always ESS. We only need to consider CD and CP norms, as these are the only norms that can be ESS and more efficient than DD norms. This is because CC (ALLC) and PP (ALLP) norms are never ESS, and DP norms are always less efficient than DD norms. We do not need to consider the other norms (DC, PC, and PD) since they are equivalent to the above norms by simply swapping the labels of G and B.

3.1. ESS condition for CD norms

First, we consider CD norms, which prescribe C against G-recipients and D against B-recipients. We derive the condition for CD norms to be ESS. In the previous work (Ohtsuki et al., 2009), the Bellman equation was derived to identify ESS norms. Here, we use a simpler approach to identify ESS norms, inspired by Murase and Hilbe (2023).

Suppose the population follows a CD norm. We consider the best response in each context, meaning the action that maximizes the donor's payoff. Here, the context refers to the recipient's reputation. For instance, we compare the expected payoffs when the donor chooses C, D, or P against a G-recipient. If the expected payoff from C is higher than that from D or P, the donor's best response against a G-recipient is C. The same comparison is made for B-recipients. If the best response against a B-recipient is D, then the resident norm is the best response to itself, making the norm ESS.

To compare the expected payoff for each action, we first calculate the expected payoff of a player currently assessed as G. In subsequent interactions, the player will be chosen as a donor with probability $1/2$ and as a recipient with the same probability. This player receives a benefit b as a recipient as long as they are assessed as G and typically has one interaction as a recipient before being chosen as a donor. When they are chosen as a donor, a new reputation is assigned, which is completely independent of the current reputation since the social norm is second-order. Thus, being assessed as G is worth b more than being assessed as B.

Now, comparing actions is straightforward. First, we compare C and D against G-recipients. In the subsequent interaction as a recipient, a cooperating donor receives a benefit b with probability $\tilde{R}(G,C)$ and a payoff of zero with probability $1-\tilde{R}(G,C)$. The total expected payoff over two rounds (one as a donor and one as a recipient) is $\tilde{R}(G,C)\,b-c$. In contrast, the expected payoff of defection is similarly expressed as $\tilde{R}(G,D)\,b$. Thus, cooperation is better than defection if and only if $\tilde{R}(G,C)\,b-c > \tilde{R}(G,D)\,b$. Similarly, we compare C and P against G-recipients, D and C against B-recipients, and D and P against B-recipients. The ESS conditions for CD norms are summarized by the following four inequalities:

$$\left[\tilde{R}(G,C)-\tilde{R}(G,D)\right]b > c \tag{2}$$

$$\left[\tilde{R}(G,C)-\tilde{R}(G,P)\right]b > c-\alpha \tag{3}$$

$$\left[\tilde{R}(B,D)-\tilde{R}(B,C)\right]b > -c \tag{4}$$

$$\left[\tilde{R}(B,D)-\tilde{R}(B,P)\right]b > -\alpha \tag{5}$$

These are simplified as the following, respectively:

$$q\xi\left[R(G,C)-R(G,D)\right]b > c \tag{6}$$

$$q\left[R(G,C)-R(G,P)\right]b > c-\alpha \tag{7}$$

$$q\xi\left[R(B,D)-R(B,C)\right]b > -c \tag{8}$$

$$q\xi\left[R(B,D)-R(B,C)\right]b+q\left[R(B,C)-R(B,P)\right]b > -\alpha \tag{9}$$

From Eqs. (6) and (7), the following conditions are necessary for a norm to be ESS:

$$R(G,C) = 1 \tag{10}$$

$$R(G,D) = 0 \tag{11}$$

$$R(G,P) = \begin{cases} 0 & \text{if } c \ge \alpha \\ \text{any} & \text{if } c < \alpha \end{cases} \tag{12}$$

$$q\xi b > c \tag{13}$$

Norms that satisfy the remaining conditions in Eqs. (8), (9), and (13) are listed in Table 2.

Table 2
Social norms that satisfy the conditions in Eqs. (8), (9), and (13) are listed. The conditions on the environmental parameters q, ξ, b, c, and α, which are additional requirements to the condition in Eq. (13), are shown in the middle column. The cooperation level calculated from Eq. (15) is displayed in the rightmost column.

$R(B,C)$	$R(B,D)$	$R(B,P)$	Condition	Cooperation level
1	1	any	any	$1-\mu$
0	1	1	$q(1-\xi)b < \alpha$	$1-\mu/\left[1-(1-2\mu)\epsilon_{DC}\right]$
0	1	0	any	$1-\mu/\left[1-(1-2\mu)\epsilon_{DC}\right]$
0	0	1	$qb < \alpha$	1/2
0	0	0	any	1/2

The cooperation level for these ESS norms is calculated. The average reputation (the fraction of G-players), denoted as h, follows this time evolution equation:

$$\dot{h} = h\tilde{R}(G,S(G))+(1-h)\,\tilde{R}(B,S(B))-h \tag{14}$$

Thus, the equilibrium reputation h^* for CD norms, which corresponds to the cooperation level, is given by

$$h^* = \frac{\tilde{R}(B,D)}{1-\tilde{R}(G,C)+\tilde{R}(B,D)} \tag{15}$$

Since $\tilde{R}(G,C) = 1-\mu$ from Eq. (10), the cooperation levels for the norms in Table 2 are calculated and presented in the rightmost column.

Table 3
Social norms that satisfy the conditions in Eqs. (22) and (23) are listed. The conditions on the environmental parameters q, ξ, b, c, and α, which are additional requirements to the condition in Eq. (13), are provided. The cooperation level calculated from Eq. (28) is displayed in the rightmost column.

$R(B,C)$	$R(B,D)$	$R(B,P)$	Condition	Cooperation level
0	0	1	$q(b+\beta)>\alpha$	$1-\mu$
0	1	1	$q(1-\xi)(b+\beta)>\alpha$	$1-\mu$
1	0	1	$c>\alpha$ and $q\xi(b+\beta)>\alpha$	$1-\mu$

Table 4
The most efficient CD and CP norms that have the widest ESS parameter regions are summarized. These norms correspond to the ones shown at the top of Table 2 and Table 3, respectively. The CD norm is stable when $q\xi b>c$. The CP norm is stable under the conditions $q\xi(b+\beta)>c$ and $q(b+\beta)>\alpha$. For both norms, the cooperation level is $1-\mu$.

$S(G)$	$S(B)$	$R(G,C)$	$R(G,D)$	$R(G,P)$	$R(B,C)$	$R(B,D)$	$R(B,P)$
C	D	1	0	0	1	1	any
C	P	1	0	0	0	0	1

Therefore, the norm at the top of Table 2 has the highest cooperation level and the widest parameter region. This norm, which is similar to Simple Standing, is the most efficient because the entry $R(B,C)=1$ does not hurt the reputation under detection error.

In summary, ESS CD norms achieve the highest cooperation level of $1-\mu$ when $q\xi b>c$. Otherwise (i.e., when $q\xi b\le c$), no CD norms are ESS.

3.2. ESS condition for CP norms

Next, we consider CP norms, where a donor cooperates with G recipients and punishes B recipients. Similar to the CD norms, we compare the actions in each context. The ESS conditions are expressed as the following four inequalities:

$$\left[\tilde{R}(G,C)-\tilde{R}(G,D)\right](b+\beta)>c \tag{16}$$

$$\left[\tilde{R}(G,C)-\tilde{R}(G,P)\right](b+\beta)>c-\alpha \tag{17}$$

$$\left[\tilde{R}(B,P)-\tilde{R}(B,C)\right](b+\beta)>\alpha-c \tag{18}$$

$$\left[\tilde{R}(B,P)-\tilde{R}(B,D)\right](b+\beta)>\alpha \tag{19}$$

Note that under CP norms, being assessed as G is worth $(b+\beta)$ more than being assessed as B, since a B recipient is punished by $-\beta$. These inequalities can be simplified as follows:

$$q\xi\left[R(G,C)-R(G,D)\right](b+\beta)>c \tag{20}$$

$$q\left[R(G,C)-R(G,P)\right](b+\beta)>c-\alpha \tag{21}$$

$$q\left[R(B,P)-R(B,C)\right](b+\beta)>\alpha-c \tag{22}$$

$$q\left\{\left[R(B,P)-R(B,C)\right]+\xi\left[R(B,C)-R(B,D)\right]\right\}(b+\beta)>\alpha \tag{23}$$

From Eqs. (20) and (21), the following conditions are necessary for a norm to be ESS:

$$R(G,C)=1 \tag{24}$$

$$R(G,D)=0 \tag{25}$$

$$R(G,P)=\begin{cases}0 & \text{if } c\ge\alpha\\ \text{any} & \text{if } c<\alpha\end{cases} \tag{26}$$

$$q\xi(b+\beta)>c \tag{27}$$

Norms that satisfy the remaining conditions from Eqs. (22) and (23) are listed in Table 3.

The equilibrium reputation h^*, which is equal to the cooperation level, is obtained from Eq. (14) as follows:

$$h^*=\frac{\tilde{R}(B,P)}{1-\tilde{R}(G,C)+\tilde{R}(B,P)} \tag{28}$$

Since $\tilde{R}(G,C)=1-\mu$ from Eq. (24) and $\tilde{R}(B,P)=1-\mu$ from Table 3, the cooperation levels for these norms are $1-\mu$.

The norm at the top of Table 3 has the widest parameter region. This norm is similar to the Stern Judging norm in that any action not prescribed by the norm always leads to a bad reputation.

In summary, ESS norms exist when $q\xi(b+\beta)>c$ and $q(b+\beta)>\alpha$. Otherwise, CP norms cannot be ESS.

The most efficient CD and CP norms are summarized in Table 4. In the following, we compare the most efficient norms and the DD norms to identify the most efficient ones among them.

3.3. The most efficient ESS norms

The equilibrium payoffs under the most efficient ESS norms obtained above are given by:

$$\pi_{CD}=(1-\mu)(b-c) \tag{29}$$

$$\pi_{CP}=(1-\mu)(b-c)-\mu(\alpha+\beta) \tag{30}$$

$$\pi_{DD}=0 \tag{31}$$

Since $\pi_{CD}>\pi_{CP}$ and $\pi_{CD}>\pi_{DD}$, CD norms are the most efficient if they exist. The comparison between CP norms and DD norms is more intricate. CP norms are more efficient than DD norms ($\pi_{CP}>\pi_{DD}$) when:

$$q>\frac{-b+c+\alpha+\beta}{b-c+\alpha+\beta} \tag{32}$$

Therefore, the most efficient ESS norms are

$$\begin{cases}\text{CD norms} & \text{when } q\xi b>c\\ \text{CP norms} & \text{when } q\xi b\le c \text{ and } q\xi(b+\beta)>c \text{ and } q(b+\beta)>\alpha \text{ and } q>\frac{-b+c+\alpha+\beta}{b-c+\alpha+\beta}\\ \text{DD norms} & \text{otherwise}\end{cases} \tag{33}$$

With perfect defection detectability ($\xi=1$), the model reproduces the baseline model (Ohtsuki et al., 2009). As derived in previous work, the most efficient norms are

$$\begin{cases}\text{CD norms} & \text{when } qb>c\\ \text{CP norms} & \text{when } qb\le c \text{ and } q(b+\beta)>c \text{ and } q(b+\beta)>\alpha \text{ and } q>\frac{-b+c+\alpha+\beta}{b-c+\alpha+\beta}\\ \text{DD norms} & \text{otherwise}\end{cases} \tag{34}$$

These regions are depicted in the top panels of Fig. 2. The regions where CP norms are the most efficient are small, and in most cases, the most efficient ESS norms are either CD norms or DD norms.

On the other hand, with perfect social resolution ($q\to 1$), the most efficient ESS norms are:

$$\begin{cases}\text{CD norms} & \text{when } \xi b>c\\ \text{CP norms} & \text{when } \xi b\le c \text{ and } \xi(b+\beta)>c \text{ and } (b+\beta)>\alpha\\ \text{DD norms} & \text{otherwise}\end{cases} \tag{35}$$

In this case, the parameter region where CP norms are the most efficient is much larger than in the baseline model, as depicted in the lower panels of Fig. 2. Furthermore, CP norms become as efficient as CD norms when $q\to 1$.

A crossover between these two limiting cases is shown in Fig. 3 to demonstrate that these behaviors are not unique to the limiting cases ($\xi=1$ or $q\to 1$). In this figure, the most efficient norms are shown for intermediate values of ξ and q while keeping the overall error rate fixed at $q\xi=0.5$. As shown in this figure, the region where CP norms are effective gradually expands as ξ decreases and q increases. Furthermore, as q increases, the cooperation level in CP norms improves and approaches that in CD norms. Thus, high social

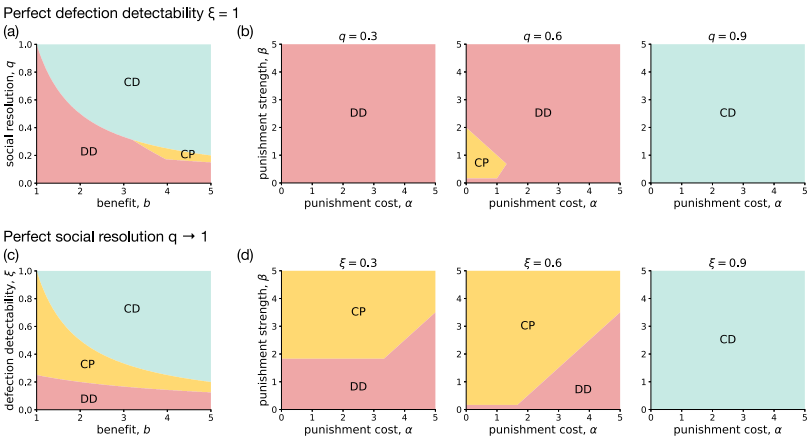


Fig. 2. The parameter regions where the most efficient ESS norms are CD norms, CP norms, and DD norms. The upper panels (a, b) show cases with perfect defection detectability ($\xi = 1$), while the lower panels (c, d) depict cases with perfect social resolution ($q \to 1$). The regions are derived from Eq. (34) for the upper panels and Eq. (35) for the lower panels. The left panels (a) and (c) display projections in the b-q and b-ξ planes, respectively, for $\alpha = 1.2$ and $\beta = 3$. The right panels (b) and (d) display projections in the α-β planes for different values of q and ξ, respectively.

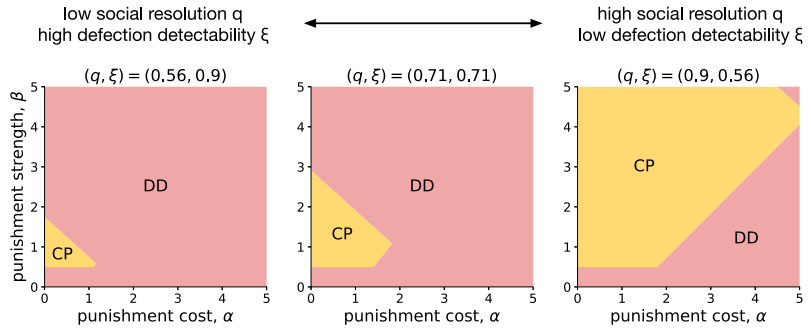


Fig. 3. The same figure as Fig. 2(b) and (d), but with intermediate values of ξ and q, while keeping $q\xi = 0.5$ and $b = 1.5$ fixed. From left to right, the values are $(q, \xi) = (5/9, 9/10)$, $(1/\sqrt{2}, 1/\sqrt{2})$, and $(9/10, 5/9)$. These regions are calculated using Eq. (33).

resolution is crucial for the effectiveness of CP norms.

The stark difference between high ξ and high q can be understood as follows. First, when both ξ and q are high, CD norms are sufficient to maintain evolutionarily stable cooperation through Simple Standing or Stern Judging norms. Punishment may be effective when the environment is noisy, either due to low ξ or low q. In these cases, the population cannot accurately identify defectors, making punishment necessary to sufficiently penalize defectors and stabilize cooperation. However, when q is low, the cooperation level is significantly reduced because the population includes individuals with bad reputations. When players punish bad recipients under CP norms, the equilibrium payoffs are significantly reduced, eventually becoming less efficient than even DD norms. In summary, while costly punishment stabilizes cooperation in noisy environments, it sacrifices the efficiency of cooperation. As a result, CP norms rarely emerge as the most efficient ESS norms, as demonstrated in Ohtsuki et al. (2009).

On the other hand, when q is sufficiently high but ξ is low, the fraction of good players h^* and the cooperation level remain high because defectors are misperceived as cooperators. As a result, the equilibrium payoffs under CP norms are as high as those under CD norms. The parameter region where CP norms are the most efficient is much larger than in the baseline model. Even though defectors are not accurately identified, cooperation can be stably maintained with a high β. The risk of strong punishment, despite its low probability, discourages players from defecting.

4. Discussion

In this study, we explored the role of costly punishment in the context of indirect reciprocity, with a particular focus on situations where the detection of defection is error-prone. Our findings challenge the conclusions of previous studies, which suggested that costly punishment offers only a narrow margin of efficiency in promoting cooperation (Ohtsuki et al., 2009). Specifically, earlier research demonstrated that while costly punishment might contribute to evolutionary stability in noisy environments, it does so at the expense of significantly reducing overall payoffs. Contrary to these earlier conclusions, our study shows that costly punishment can indeed be beneficial under certain conditions, particularly when other types of errors are present.

We focused on scenarios where defection might be misperceived as cooperation——a situation that previous models have not adequately addressed. These scenarios are realistic since defectors have a natural incentive to hide their defection and deceive others into perceiving their actions as cooperative. Our results indicate that under these circumstances, costly punishment can facilitate evolutionarily stable cooperation without severely diminishing the overall level of cooperation. This finding suggests that costly punishment plays a more effective role in indirect reciprocity than previously thought, especially in environments where accurately identifying defection is challenging.

From a mathematical perspective, the key to the effectiveness of costly punishment in our model lies in the asymmetric nature of the errors involved. In traditional models of indirect reciprocity, errors are typically induced symmetrically; that is, the probability of a good action being misassessed as bad is equal to the probability of a bad action being misassessed as good. However, in our model, these errors are induced asymmetrically, with the likelihood of misperceiving defection as cooperation differing from the likelihood of misperceiving cooperation as defection. This asymmetry is critical because it allows populations to maintain a high level of cooperation even in the presence of noise, making CP norms significantly more efficient compared

to the symmetric error model. Even though defection is harder to detect under the asymmetric errors, the population can still effectively suppress free-riders via strong punishment, thereby sustaining cooperation. We may interpret such an asymmetry in errors as a custom that the society intentionally implemented. Sometimes the identification of actions may be imperfect due to noise or cost. It is beneficial to assess uncertain players as good rather than randomly assigning them a good or bad label. By doing so, the population can maintain a high level of cooperation. To compensate for the risk of misidentifying defectors as cooperators, the society needs to implement a strong punishment system to deter potential defectors from exploitation. This is how costly punishment can be effective in promoting cooperation in indirect reciprocity even if it incurs some cost. Future studies on models with uncertain states like those in Nakamura and Masuda (2011), or on models that account for the cost of reputation sharing (Yamamoto et al., 2024), may provide further insights.

While we focused on the errors in perceiving defection, other types of errors could also be considered (Ohtsuki et al., 2009; Nakamura and Masuda, 2011; Sasaki et al., 2017; Radzvilavicius et al., 2019). A commonly studied type is execution error, where the donor's intended action may not be carried out correctly. Another potential error is the misperception of the donor's cooperation. When the benefits of cooperation are not immediately recognizable, cooperative actions may be misperceived as defection. Additionally, errors involving punishment actions could also be explored. Although we did not examine these errors comprehensively in this paper, costly punishment is likely to be effective when such errors lead to higher levels of cooperation. As discussed earlier, costly punishment becomes inefficient when it must be applied frequently. For example, frequent execution errors make costly punishment ineffective (see Appendix A for an analysis of the model with execution errors). As a general guideline, costly punishment is most effective when it is rarely executed. Punishment deters defection primarily through the threat of punishment rather than its frequent application.

While our findings provide new insights into the role of costly punishment in indirect reciprocity, several open questions remain for future research. In this paper, we focused on ESS analysis to examine the stability of cooperation once social norms are established. However, this represents only one part of the evolutionary process. An interesting direction for future work is to investigate how social norms incorporating costly punishment can initially evolve. This is a complex problem, as costly punishment often leads to inefficiencies when cooperators and free-riders coexist. We surmise that costly punishment may not easily evolve unless a highly cooperative state is initially established. Further research is needed to elucidate the evolutionary dynamics and conditions under which cooperation sustained by costly punishment emerges and stabilizes.

In this study, we focused on second-order social norms, which represent the simplest class of norms that can sustain evolutionarily stable cooperation. Future studies should explore more complex strategies (Santos et al., 2018), such as third-order social norms (Ohtsuki and Iwasa, 2004), stochastic norms (Murase and Hilbe, 2023; Schmid et al., 2021a,b), and norms involving non-binary reputations (Murase et al., 2022; Schmid et al., 2023; Lee et al., 2021). For example, in the realm of third-order social norms, the "secondary sixteen" norms have been shown to promote cooperation (Murase and Hilbe, 2023), and it would be intriguing to investigate the effectiveness of costly punishment within these norms.

Another promising avenue for future research is the study of dual-reputation updates (Murase and Hilbe, 2023), where not only the donor's reputation but also the recipient's reputation is updated. Previous research suggests that dual-reputation updates make norms more error-tolerant and thus more efficient, albeit at the cost of requiring a higher benefit-to-cost ratio to maintain cooperation. It would be interesting to examine whether costly punishment could reduce this required ratio while preserving the efficiency of dual-reputation systems.

Additionally, the private assessment models, in which players' assessments of others may not always be synchronized, presents another interesting direction for future studies (Hilbe et al., 2018; Murase and Hilbe, 2024b). In such models, costly punishment may be particularly effective because it could help distinguish free-riders from those who engage in justified punishment. A major open question in this area is how costly punishment and social norms can co-evolve initially (Murase and Hilbe, 2024a). We hypothesize that it may evolve in situations where observing actions is costly and error-prone, and punishment emerges as a way to suppress free-riding in this challenging environment.

In conclusion, our study demonstrates that costly punishment can be a valuable mechanism in sustaining cooperation in indirect reciprocity, particularly in scenarios where errors in detecting defection are present. These findings open up new avenues for exploring more complex social norms and assessment models, potentially leading to a deeper understanding of the mechanisms that underpin cooperation in human societies.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used chatGPT in order to improve the readability. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Execution errors

Here, we present an analysis of the model with execution errors, where the donor's intended action may not be carried out correctly. Following previous studies, we assume that the donor's intended cooperation is successfully executed with probability $1-\mu_e$, while defection is incorrectly executed with probability μ_e. All other actions, D and P, are assumed to be executed correctly.

Since intended cooperation may be executed erroneously, we define the effective benefit and cost as follows:

$$b^\dagger = (1-\mu_e)b \tag{A.1}$$

$$c^\dagger = (1-\mu_e)c \tag{A.2}$$

We also introduce an effective assessment rule $\tilde{R}^\dagger(X,C)$, which represents the probability of assigning a good reputation to a donor who intends to cooperate:

$$\tilde{R}^\dagger(X,C) = \left(1-\mu_e\right)\tilde{R}(X,C) + \mu_e\tilde{R}(X,D) \tag{A.3}$$

Since the other actions are executed correctly, the effective assessment rules are remain identical to the original ones:

$$\tilde{R}^\dagger(X,D) = \tilde{R}(X,D) \tag{A.4}$$

$$\tilde{R}^\dagger(X,P) = \tilde{R}(X,P) \tag{A.5}$$

Using these effective assessment rules, as well as the effective benefit and cost, we can derive the ESS conditions for CD norms and CP norms under execution errors. Let us first extend the analysis in Section 3.1, which addresses the ESS condition for CD norms. Eqs. (2)–(5) are modified as follows:

$$\left[\tilde{R}^{\dagger}(G,C)-\tilde{R}^{\dagger}(G,D)\right]b^{\dagger}>c^{\dagger} \tag{A.6}$$

$$\left[\tilde{R}^{\dagger}(G,C)-\tilde{R}^{\dagger}(G,P)\right]b^{\dagger}>c^{\dagger}-\alpha \tag{A.7}$$

$$\left[\tilde{R}^{\dagger}(B,D)-\tilde{R}^{\dagger}(B,C)\right]b^{\dagger}>-c^{\dagger} \tag{A.8}$$

$$\left[\tilde{R}^{\dagger}(B,D)-\tilde{R}^{\dagger}(B,P)\right]b^{\dagger}>-\alpha \tag{A.9}$$

The following analysis proceeds analogously to the original one, yielding the ESS conditions. While we omit the detailed calculations here, it is important to emphasize that small execution errors do not qualitatively affect the ESS conditions.

A similar analysis can be conducted for CP norms by replacing the condition in Section 3.2 with the effective benefit, cost, and assessment rules. As before, we omit the detailed calculations, but the ESS conditions for CP norms under execution errors are derived in the same manner as the original ones.

The cooperation levels for the most efficient CD norms and CP norms under execution errors can be calculated in the same way as the previous ones. By substituting the effective assessment rules into Eqs. (15) and (28), the corresponding cooperation levels are derived. One of the most efficient CD norms is $R(G,C)=1$, $R(G,D)=0$, $R(G,P)=0$, $R(B,C)=1$, $R(B,D)=1$, and $R(B,P)=0$, whose cooperation level is

$$p_c=(1-\mu_e)h^{*}=\frac{(1-\mu_e)(1-\mu)}{1+q\xi\mu_e} \tag{A.10}$$

Similarly, one of the most efficient CP norms is $R(G,C)=1$, $R(G,D)=0$, $R(G,P)=0$, $R(B,C)=0$, $R(B,D)=0$, and $R(B,P)=1$. The cooperation level is identical to Eq. (A.10).

As μ_e increases, the cooperation level declines, making these norms less efficient than the original ones. Consequently, the parameter region where CP norms are the most efficient becomes smaller, and the most efficient norms are more likely to be CD norms or DD norms.

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